

UNIT-III - TRIGONOMETRY (18 MARKS IN GTU)

* Convert Radian into degree form
multiply by $\frac{\pi}{180}$.

* Convert degree into Radian form
multiply by $\frac{180}{\pi}$.

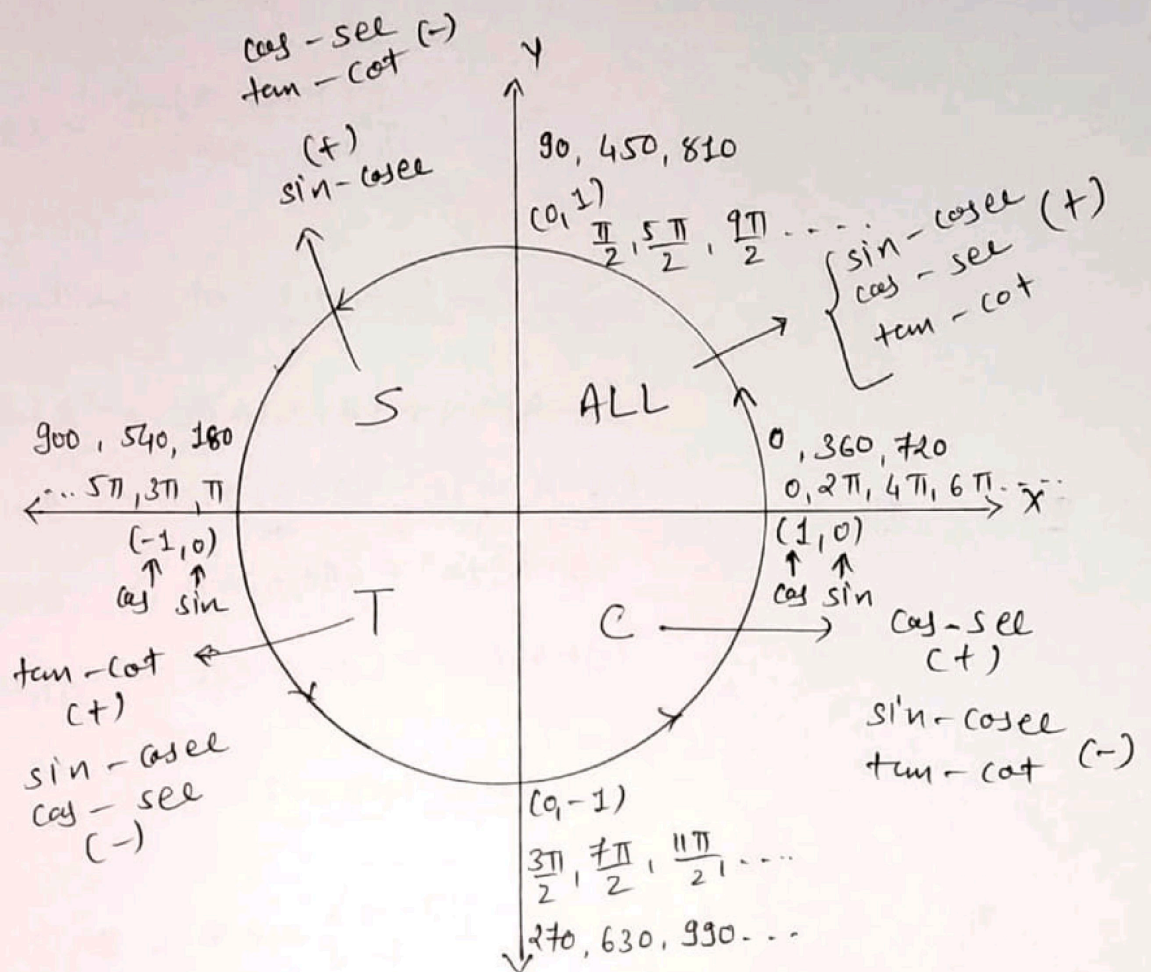
* Values of Trigonometric Function

Trigonometric Function \ Angle	0(0°)	$\frac{\pi}{6}$ (30°)	$\frac{\pi}{4}$ (45°)	$\frac{\pi}{3}$ (60°)	$\frac{\pi}{2}$ (90°)
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$2/\sqrt{3}$	1
$\sec \theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$1/\sqrt{3}$	0

* $\sin^2 \theta + \cos^2 \theta = 1$

$\sec^2 \theta - \tan^2 \theta = 1$

$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$



→ Function change at angle $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots$

In short, Function change on y-axis's angles.

$$\ast \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\rightarrow \cot(A+B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$$

$$\cdot \cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

$\rightarrow$ Multiplication to Summation

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$\rightarrow$ Summation to Multiplication

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$\rightarrow$ Multiple - submultiple Angle

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$\tan 2\theta = \frac{2\tan \theta}{1 - \tan^2 \theta}$$

$$\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$$

$$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$$

$$\tan 3\theta = \frac{3\tan \theta - (\tan \theta)^3}{1 - 3\tan^2 \theta}$$

→ Inverse Trigonometric Function :-

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right), \text{ if } xy < 1$$

$$= \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), \text{ if } xy > 1$$

$$= \frac{\pi}{2}, \text{ if } xy = 1$$

$$\sin^{-1} x = \cos^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{x}{\sqrt{1-x^2}}, \quad 0 < x < 1$$

$$\cos^{-1} x = \sin^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{\sqrt{1-x^2}}{x}, \quad 0 < x < 1$$

$$\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1+x^2}} = \cos^{-1} \frac{1}{\sqrt{1+x^2}}, \quad x > 0$$

* Question for 1 Marks

1. Convert into degree.

$$\frac{3\pi}{4}, \quad \frac{7\pi}{6}$$

$$\frac{3\pi}{4} \times \frac{180}{\pi} = 135^\circ, \quad \frac{7\pi}{6} \times \frac{180}{\pi} = 210^\circ$$

2. Convert into Radian. 150° , 20°

$$150^\circ = 150 \times \frac{\pi}{180} = \frac{5\pi}{6}$$

$$20^\circ = 20 \times \frac{\pi}{180} = \frac{\pi}{9}$$

3. If $\cos \theta = \frac{\sqrt{3}}{2}$, $\sin \theta = -\frac{1}{2}$ then θ lies in
4th quadrant because $\cos$ is positive and
 $\sin$ is negative.

4. $\cos 90^\circ \times \cos 60^\circ \times \sin 30^\circ$

$$= 0 \times \frac{1}{2} \times \frac{1}{2}$$

$$= \boxed{0}$$

5. $\tan 225^\circ$

$$= \tan (180 + 45) = \tan 45 = \boxed{1}$$

$$6. \cot(-30^\circ)$$

$$= -\cot 30^\circ = \boxed{-\sqrt{3}}$$

$$7. \sin^2 57^\circ + \sin^2 33^\circ$$

$$= \sin^2 57^\circ + \sin^2 (90^\circ - 57^\circ)$$

$$= \sin^2 57^\circ + \cos^2 57^\circ$$

$$= \boxed{1}$$

$$8. \cos^2 30^\circ + \cos^2 60^\circ$$

$$= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{3}{4} + \frac{1}{4} = \frac{3+1}{4} = \frac{4}{4} = \boxed{1}$$

$$9. \text{ If } A+B+C = \pi \text{ then } \cos\left(\frac{B+C}{2}\right) \text{ is}$$

$$\therefore B+C = \pi - A$$

$$\therefore \frac{B+C}{2} = \frac{\pi - A}{2}$$

$$\text{Now, } \cos\left(\frac{B+C}{2}\right) = \cos\left(\frac{\pi - A}{2}\right)$$

$$= \cos\left(\frac{\pi}{2} - \frac{A}{2}\right)$$

$$= \sin \frac{A}{2}$$

$$10. \cos x + \cos(\pi - x) + \cos(2\pi - x) + \cos(3\pi - x)$$

$$= \cos x + (-\cos x) + \cos x + (-\cos x)$$

$$= \cancel{\cos x} - \cancel{\cos x} + \cancel{\cos x} - \cancel{\cos x}$$

$$= 0$$

$$11. \text{ period of } \sin x = \frac{2\pi}{2} = \pi.$$

$$12. \text{ period of } \tan\left(3x + \frac{\pi}{6}\right) = \frac{\pi}{3}$$

$$13. \sin^{-1} \frac{1}{2} = \frac{\pi}{6}.$$

$$14. \cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1} \frac{1}{2} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$15. \cos\left(\cos^{-1} \frac{2}{3}\right) = \frac{2}{3}$$

$$16. \sin(\tan^{-1} p + \cot^{-1} p) \\ = \sin \frac{\pi}{2} = \boxed{1}$$

* Question for 3 Marks:-

$$1. \text{ Prove that } \tan 225^\circ \times \cot 405^\circ + \tan 765^\circ \times \cot 675^\circ = 0.$$

$$\text{L.H.S} = \tan 225^\circ \times \cot 405^\circ + \tan 765^\circ \times \cot 675^\circ$$

$$= \tan(180+45) \times \cot(360+45) + \tan(720+45) + \cot(630+45)$$

$$= \tan 45 \times \cot 45 + \tan 45 \times (-\tan 45)$$

$$= 1 - (1)(1)$$

$$= 1 - 1$$

$$= 0$$

$$= \text{R.H.S}$$

2. Prove that $\cos \frac{19\pi}{6} \sin \frac{17\pi}{6} - \sin \frac{11\pi}{6} \cos \frac{13\pi}{6} = 0$.

$$\text{L.H.S} := \cos \frac{19\pi}{6} \sin \frac{17\pi}{6} - \sin \frac{11\pi}{6} \cos \frac{13\pi}{6}$$

$$= \cos\left(3\pi + \frac{\pi}{6}\right) \sin\left(3\pi - \frac{\pi}{6}\right) - \sin\left(2\pi - \frac{\pi}{6}\right) \cos\left(2\pi + \frac{\pi}{6}\right)$$

$$= \cancel{+ \sin} - \cos \frac{\pi}{6} \sin \frac{\pi}{6} - (-\sin \frac{\pi}{6}) \cos \frac{\pi}{6}$$

$$= -\cos \frac{\pi}{6} \sin \frac{\pi}{6} + \sin \frac{\pi}{6} \cos \frac{\pi}{6}$$

$$= -\frac{\sqrt{3}}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2}$$

$$= -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4}$$

$$= 0$$

$$= \text{R.H.S}$$

3. Prove that $\sin^2 \frac{\pi}{4} + \sin^2 \frac{3\pi}{4} + \sin^2 \frac{5\pi}{4} + \sin^2 \frac{7\pi}{4} = 2$.

$$\text{L.H.S} = \sin^2 \frac{\pi}{4} + \sin^2 \frac{3\pi}{4} + \sin^2 \frac{5\pi}{4} + \sin^2 \frac{7\pi}{4}$$

$$= \sin^2 \frac{\pi}{4} + \sin^2\left(\pi - \frac{\pi}{4}\right) + \sin^2\left(\pi + \frac{\pi}{4}\right) + \sin^2\left(2\pi - \frac{\pi}{4}\right)$$

$$= \sin^2 \frac{\pi}{4} + \sin^2 \frac{\pi}{4} + \sin^2 \frac{\pi}{4} + \sin^2 \frac{\pi}{4}$$

$$= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2$$

$$= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 2 = \text{R.H.S.}$$

4. Prove that $\tan \frac{\pi}{20} \tan \frac{3\pi}{20} \tan \frac{5\pi}{20} \tan \frac{7\pi}{20} \tan \frac{9\pi}{20} = 1$

$$\rightarrow \tan \frac{9\pi}{20} = \tan \left(\frac{\pi}{2} - \frac{\pi}{20} \right) = \cot \frac{\pi}{20}$$

$$\tan \frac{7\pi}{20} = \tan \left(\frac{\pi}{2} - \frac{3\pi}{20} \right) = \cot \frac{3\pi}{20}$$

$$\tan \frac{5\pi}{20} = \tan \frac{\pi}{4} = 1$$

$$\text{L.H.S} = \tan \frac{\pi}{20} \tan \frac{3\pi}{20} \tan \frac{5\pi}{20} \tan \frac{7\pi}{20} \tan \frac{9\pi}{20}$$

$$= \tan \frac{\pi}{20} \cdot \tan \frac{3\pi}{20} \cdot 1 \cdot \cot \frac{3\pi}{20} \cdot \cot \frac{\pi}{20}$$

$$= \left(\tan \frac{\pi}{20} \cot \frac{\pi}{20} \right) \cdot \left(\tan \frac{3\pi}{20} \cot \frac{3\pi}{20} \right)$$

$$= (1) (1)$$

$$= 1$$

$$= \text{R.H.S.}$$

5. P. T. $\sin(A+B) \cdot \sin(A-B) = \sin^2 A - \sin^2 B$
 $= \cos^2 B - \cos^2 A$

$$\text{L.H.S} = \sin(A+B) \cdot \sin(A-B)$$

$$= (\sin A \cos B + \cos A \sin B) (\sin A \cos B - \cos A \sin B)$$

$$= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B$$

$$= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B$$

$$(\sin^2 A + \cos^2 A = 1)$$

$$= \sin^2 A - \cancel{\sin^2 A \sin^2 B} - \sin^2 B + \cancel{\sin^2 A \sin^2 B}$$

$$= \sin^2 A - \sin^2 B = \text{R.H.S}$$

$$= \sin^2 A - \sin^2 B$$

$$= 1 - \cos^2 A - (1 - \cos^2 B)$$

$$= \cancel{1} - \cos^2 A - \cancel{1} + \cos^2 B$$

$$= \cos^2 B - \cos^2 A$$

$$= R.H.S$$

6. Prove that $\cos A \cdot \sin(B-C) + \cos B \cdot \sin(C-A) + \cos C \cdot \sin(A-B) = 0$

$$\cos A \cdot \sin(B-C)$$

$$= \cos A (\sin B \cos C - \cos B \sin C)$$

$$= \cos A \sin B \cos C - \cos A \cos B \sin C$$

$$\text{Similarly, } \cos B \sin(C-A)$$

$$= \cos B \sin C \cos A - \cos B \sin A \cos C$$

$$= \cos A \cos B \sin C - \sin A \cos B \cos C$$

$$\rightarrow \cos C \sin(A-B)$$

$$\sin A \cos B \cos C - \cos A \sin B \cos C$$

$$L.H.S = \cos A \cdot \sin(B-C) + \cos B \sin(C-A) + \cos C \sin(A-B)$$

$$= \cancel{\cos A} \sin B \cos C - \cos A \cancel{\cos B} \sin C$$

$$+ \cos A \cancel{\cos B} \sin C - \sin A \cancel{\cos B} \cos C$$

$$+ \sin A \cancel{\cos B} \cos C - \cos A \cancel{\sin B} \cos C$$

$$= 0$$

$$= R.H.S$$

7. prove that $\tan 55^\circ = \frac{\cos 10^\circ + \sin 10^\circ}{\cos 10^\circ - \sin 10^\circ}$

$$\text{L.H.S} = \tan 55^\circ$$

$$= \tan(45^\circ + 10^\circ)$$

$$= \frac{\tan 45^\circ + \tan 10^\circ}{1 - \tan 45^\circ \tan 10^\circ}$$

$$= \frac{1 + \tan 10^\circ}{1 - \tan 10^\circ} \quad (\because \tan 45^\circ = 1)$$

$$= \frac{1 + \frac{\sin 10^\circ}{\cos 10^\circ}}{1 - \frac{\sin 10^\circ}{\cos 10^\circ}}$$

$$= \frac{\cos 10^\circ + \sin 10^\circ}{\cos 10^\circ - \sin 10^\circ} = \text{R.H.S.}$$

8. prove that $\frac{\sin 79^\circ + \sin 19^\circ}{\cos 19^\circ - \cos 79^\circ} = \sqrt{3}$

$$\text{L.H.S} = \frac{\sin 79^\circ + \sin 19^\circ}{\cos 19^\circ - \cos 79^\circ}$$

$$= \frac{2 \sin\left(\frac{79^\circ + 19^\circ}{2}\right) \cos\left(\frac{79^\circ - 19^\circ}{2}\right)}{-2 \sin\left(\frac{19^\circ + 79^\circ}{2}\right) \sin\left(\frac{19^\circ - 79^\circ}{2}\right)}$$

$$= \frac{2 \sin\left(\frac{98^\circ}{2}\right) \cos\left(\frac{60^\circ}{2}\right)}{-2 \sin\left(\frac{98^\circ}{2}\right) \sin\left(-\frac{60^\circ}{2}\right)}$$

$$\begin{aligned} S + S &= 2SC \\ C - C &= -2SS \end{aligned}$$

$$= \frac{2 \sin 49^\circ \cos 30^\circ}{-2 \sin 49^\circ \sin(-30^\circ)}$$

$$= \frac{\cos 30^\circ}{\sin 30^\circ} \quad (\because \sin(-\theta) = -\sin \theta)$$

$$= \cot 30^\circ$$

$$= \sqrt{3}$$

$$= \text{R.H.S.}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

9. P.T. $(1 + \tan 25^\circ)(1 + \tan 20^\circ) = 2$

~~1.30 (24 tan 25° 19 + tan 20°)~~

$$\rightarrow \tan 45^\circ = \tan(25^\circ + 20^\circ)$$

$$\therefore 1 = \frac{\tan 25^\circ + \tan 20^\circ}{1 - \tan 25^\circ \tan 20^\circ}$$

$$\therefore 1 - \tan 25^\circ \tan 20^\circ = \tan 25^\circ + \tan 20^\circ$$

$$\therefore \tan 25^\circ + \tan 20^\circ + \tan 25^\circ \tan 20^\circ = 1$$

Adding 1 on both side:

$$\therefore 1 + \tan 25^\circ + \tan 20^\circ + \tan 25^\circ \tan 20^\circ = 1 + 1$$

$$\therefore 1(1 + \tan 25^\circ) + \tan 20^\circ(1 + \tan 25^\circ) = 2$$

$$\therefore (1 + \tan 25^\circ)(1 + \tan 20^\circ) = 2$$

10. If $\tan x = \frac{5}{6}$ and $\tan y = \frac{1}{11}$ then P.T.

$$x + y = \frac{\pi}{4}$$

$$\rightarrow \tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$= \frac{5/6 + 1/11}{1 - 5/6 \cdot 1/11}$$

$$= \frac{\frac{55+6}{66}}{\frac{66-5}{66}}$$

$$= \frac{61/66}{61/66}$$

$$\therefore \tan(x+y) = 1 > 0 \quad \text{--- CD}$$

$$\therefore 0 < x < \frac{\pi}{2} \text{ \& } 0 < y < \frac{\pi}{2}$$

$$\therefore 0 < x+y < \pi$$

$$\therefore x+y = \tan^{-1} 1$$

$$\therefore \boxed{x+y = \frac{\pi}{4}}$$

11. P. T. $\sin 3A = 3\sin A - 4\sin^3 A$

$$\text{L.H.S} = \sin 3A$$

$$= \sin(2A+A)$$

$$= \sin 2A \cos A + \cos 2A \sin A$$

$$= \underline{2\sin A \cos A} \cos A + \underline{(1-2\sin^2 A)} \sin A$$

$$= 2\sin A \cos^2 A + \sin A - 2\sin^3 A$$

$$= 2\sin A (1-\sin^2 A) + \sin A - 2\sin^3 A$$

$$= \underline{2\sin A} - \underline{2\sin^3 A} + \underline{\sin A} - \underline{2\sin^3 A}$$

$$= 3\sin A - 4\sin^3 A$$

$$= \text{R.H.S.}$$

$$12. \text{ P.T. } \cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$$

$$\text{L.H.S} = \cos 20^\circ \cos 40^\circ \cos 80^\circ$$

$$= \frac{1}{2} \cos 20^\circ [2 \cos 80^\circ \cos 40^\circ]$$

$$\because 2 \cos C \cos = (\cos(C+))$$

$$= \frac{1}{2} \cos 20^\circ [\cos(80+40) + \cos(80-40)]$$

$$= \frac{1}{2} \cos 20^\circ [\cos 120 + \cos 40]$$

$$= \frac{1}{2} \cos 20^\circ [\cos(180-60) + \cos 40]$$

$$= \frac{1}{2} \cos 20^\circ [-\cos 60 + \cos 40]$$

$$= \frac{1}{2} \cos 20^\circ \left(-\frac{1}{2}\right) + \frac{1}{2} \cos 40^\circ \cos 20^\circ$$

$$= -\frac{1}{4} \cos 20^\circ + \frac{1}{2} \cdot \frac{1}{2} [2 \cos 40^\circ \cos 20^\circ]$$

$$= -\frac{1}{4} \cos 20^\circ + \frac{1}{4} [\cos(40+20) + \cos(40-20)]$$

$$= -\frac{1}{4} \cos 20^\circ + \frac{1}{4} [\cos 60 + \cos 20]$$

$$= -\frac{1}{4} \cos 20^\circ + \frac{1}{4} \left(\frac{1}{2} + \cos 20^\circ\right)$$

$$= -\frac{1}{4} \cancel{\cos 20^\circ} + \frac{1}{8} + \frac{1}{4} \cancel{\cos 20^\circ}$$

$$= \frac{1}{8}$$

$$= \text{R.H.S}$$

13. Prove that $\frac{\sin 19 + \cos 11}{\cos 19 - \sin 11} = \sqrt{3}$

$$\text{L.H.S.} = \frac{\sin 19 + \cos 11}{\cos 19 - \sin 11}$$

$$= \frac{\sin 19 + \cos (90 - 79)}{\cos 19 - \sin (90 - 79)}$$

$$= \frac{\sin 19 + \sin 79}{\cos 19 - \cos 79}$$

$$= \frac{2 \sin \left(\frac{19 + 79}{2} \right) \cos \left(\frac{19 - 79}{2} \right)}{-2 \sin \left(\frac{19 + 79}{2} \right) \sin \left(\frac{19 - 79}{2} \right)}$$

$$(\because S + S = 2SC \\ C - C = -2SS)$$

$$= \frac{2 \sin 49 \cos (-30)}{-2 \sin 49 \sin (-30)}$$

$$= \frac{\cos 30}{\sin 30} \quad (\because \cos(-\theta) = \cos \theta \\ \sin(-\theta) = -\sin \theta)$$

$$= \cot 30^\circ$$

$$= \sqrt{3}$$

$$= \text{R.H.S.}$$

14. Prove that $\sin 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{(1 + \tan^2 \theta)^2}$

$$\text{L.H.S.} = \sin 4\theta$$

$$= 2 \sin 2\theta \cos 2\theta$$

$$= 2 \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right)$$

$$= \frac{4 \tan \theta (1 - \tan^2 \theta)}{(1 + \tan^2 \theta)^2}$$

$$= R.H.S$$

15. Prove that $\frac{1 + \sin 2A - \cos 2A}{1 + \sin 2A + \cos 2A} = \tan A$

$$L.H.S = \frac{1 + \sin 2A - \cos 2A}{1 + \sin 2A + \cos 2A}$$

$$= \frac{(1 - \cos 2A) + \sin 2A}{(1 + \cos 2A) + \sin 2A}$$

$$= \frac{2 \sin^2 A + 2 \sin A \cos A}{2 \cos^2 A + 2 \sin A \cos A}$$

$$= \frac{2 \sin A (\sin A + \cos A)}{2 \cos A (\cos A + \sin A)}$$

$$= \tan A$$

$$= R.H.S.$$

16. Prove that $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$

$$L.H.S = \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}}\right)$$

$$\left(x \cdot y = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} < 1\right)$$

$$= \tan^{-1}\left(\frac{\frac{3+2}{6}}{\frac{6-1}{6}}\right)$$

$$= \tan^{-1}\left(\frac{5}{5}\right) = \tan^{-1}1 = \frac{\pi}{4} = R.H.S$$

* Question for 4 Marks :-

1. Prove that $\frac{\sin(\theta - \frac{\pi}{2})}{\cos(\theta - \pi)} + \frac{\tan(\frac{\pi}{2} - \theta)}{\cot(\pi - \theta)} + \frac{\operatorname{cosec}(\frac{\pi}{2} + \theta)}{\sec(\pi + \theta)} = -1$

$$\begin{aligned} \text{L.H.S} &= \frac{\sin(\theta - \frac{\pi}{2})}{\cos(\theta - \pi)} + \frac{\tan(\frac{\pi}{2} - \theta)}{\cot(\pi - \theta)} + \frac{\operatorname{cosec}(\frac{\pi}{2} + \theta)}{\sec(\pi + \theta)} \\ &= \frac{\sin[-(\frac{\pi}{2} - \theta)]}{\cos[-(\pi - \theta)]} + \frac{\tan(\frac{\pi}{2} - \theta)}{\cot(\pi - \theta)} + \frac{\operatorname{cosec}(\frac{\pi}{2} + \theta)}{\sec(\pi + \theta)} \\ &= \frac{-\sin(\frac{\pi}{2} - \theta)}{\cos(\pi - \theta)} + \frac{\tan(\frac{\pi}{2} - \theta)}{\cot(\pi - \theta)} + \frac{\operatorname{cosec}(\frac{\pi}{2} + \theta)}{\sec(\pi + \theta)} \\ &= \frac{-\cos\theta}{-\cos\theta} + \frac{\cot\theta}{-\cot\theta} + \frac{\sec\theta}{-\sec\theta} \\ &= 1 - 1 - 1 \\ &= -1 \\ &= \text{R.H.S} \end{aligned}$$

2. Prove that $\frac{\sin\theta + \sin 2\theta + \sin 4\theta + \sin 8\theta}{\cos\theta + \cos 2\theta + \cos 4\theta + \cos 8\theta} = \tan 3\theta$

$$\begin{aligned} \text{L.H.S} &= \frac{\sin\theta + \sin 2\theta + \sin 4\theta + \sin 8\theta}{\cos\theta + \cos 2\theta + \cos 4\theta + \cos 8\theta} \\ &= \frac{\sin 8\theta + \sin\theta + \sin 4\theta + \sin 2\theta}{\cos 8\theta + \cos\theta + \cos 4\theta + \cos 2\theta} \\ &= \frac{2\sin(\frac{8\theta + \theta}{2})\cos(\frac{8\theta - \theta}{2}) + 2\sin(\frac{4\theta + 2\theta}{2})\cos(\frac{4\theta - 2\theta}{2})}{2\cos(\frac{8\theta + \theta}{2})\cos(\frac{8\theta - \theta}{2}) + 2\cos(\frac{4\theta + 2\theta}{2})\cos(\frac{4\theta - 2\theta}{2})} \end{aligned}$$

$$= \frac{2 \sin 3\theta \cos 2\theta + 2 \sin 3\theta \cos \theta}{2 \cos 3\theta \cos 2\theta + 2 \cos 3\theta \cos \theta}$$

$$= \frac{2 \sin 3\theta (\cancel{\cos 2\theta} + \cos \theta)}{2 \cos 3\theta (\cancel{\cos 2\theta} + \cos \theta)}$$

$$= \tan 3\theta$$

$$= \text{R. H. S.}$$

3. * P. T. $\sin(180^\circ - \theta) \cos(\theta - 90^\circ) \cot(180^\circ - \theta) + \cos(360^\circ + \theta) \csc(180^\circ - \theta) \cot(90^\circ - \theta) = \sin^2 \theta.$

$$\text{L.H.S} = \sin(180^\circ - \theta) \cos(\theta - 90^\circ) \cot(180^\circ - \theta) + \cos(360^\circ + \theta) \csc(180^\circ - \theta) \cot(90^\circ - \theta)$$

$$= \sin(\pi - \theta) \cos(\theta - 90^\circ) \cot(\pi - \theta) + \cos(2\pi + \theta) \csc(\pi - \theta) \cot\left(\frac{\pi}{2} - \theta\right)$$

$$= (\sin \theta) (\cos \theta) (-\cot \theta) + (\cos \theta) (\csc \theta) (\tan \theta)$$

$$= \sin \theta \cdot \cos \theta \left(-\frac{\cos \theta}{\sin \theta}\right) + \cos \theta \left(\frac{1}{\sin \theta}\right) \left(\frac{\sin \theta}{\cos \theta}\right)$$

$$= -\cos^2 \theta + 1$$

$$= 1 - \cos^2 \theta$$

$$= \sin^2 \theta$$

$$= \text{R. H. S.}$$

4. P. T. $\frac{\tan(\pi - \theta)}{\tan(\pi + \theta)} \times \frac{\cot(\pi + \theta)}{\cot(\pi - \theta)} \times \frac{\tan(2\pi + \theta)}{\cot(2\pi - \theta)} = -\tan^2 \theta$

$$\text{L.H.S} = \frac{\tan(\pi-\theta)}{\tan(\pi+\theta)} \times \frac{\cot(\pi+\theta)}{\cot(\pi-\theta)} \times \frac{\tan(2\pi+\theta)}{\cot(2\pi-\theta)}$$

$$= \frac{-\cancel{\tan\theta}}{\tan\theta} \times \frac{\cancel{\cot\theta}}{-\cot\theta} \times \frac{\tan\theta}{-\cot\theta}$$

$$= (-1) \times (-1) \times (\tan\theta) (-\tan\theta)$$

$$= -\tan^2\theta$$

$$= \text{R.H.S.}$$

5. If A, B, C & D are angles of cyclic quadrilateral then P.T. $\cos A + \cos B + \cos C + \cos D = 0$.

→ $\square ABCD$ is a cyclic quadrilateral.

In cyclic quadrilateral sum of opposite angle is 180°

$$\therefore m\angle A + m\angle C = 180^\circ \quad \text{--- (1)}$$

$$m\angle B + m\angle D = 180^\circ \quad \text{--- (2)}$$

$$\text{Now, From (1)} \quad A + C = \pi$$

$$\therefore C = \pi - A$$

$$\therefore \cos C = \cos(\pi - A)$$

$$\therefore \cos C = -\cos A \quad \text{--- (3)}$$

$$\text{From (2), } B + D = \pi$$

$$\therefore D = \pi - B$$

$$\therefore \cos D = \cos(\pi - B)$$

$$\therefore \cos D = -\cos B \quad \text{--- (4)}$$

$$\text{L.H.S} = \cos A + \cos B + \cos C + \cos D$$

$$= \cancel{\cos A} + \cos B - \cancel{\cos A} - \cos B \quad (\because \text{by (3) \& (4)})$$

$$= 0 = \text{R.H.S.}$$

6. For any $\triangle ABC$ P.T. $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$

→ we know that for any $\triangle ABC$,

$$A + B + C = \pi$$

$$\therefore A + B = \pi - C$$

$$\therefore \tan(A + B) = \tan(\pi - C)$$

$$\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\therefore \tan A + \tan B = -\tan C (1 - \tan A \tan B)$$

$$\therefore \tan A + \tan B = -\tan C + \tan A \tan B \tan C$$

$$\therefore \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

7. Prove that $4 \sin 2A \sin(60^\circ + 2A) \sin(60^\circ - 2A) = \sin 6A$

$$\text{L.H.S} = 4 \sin 2A \sin(60^\circ + 2A) \sin(60^\circ - 2A)$$

$$= -2 \sin 2A [-2 \sin(60^\circ + 2A) \sin(60^\circ - 2A)]$$

$$= -2 \sin 2A [\cos(60^\circ + 2A + 60^\circ - 2A) - \cos(60^\circ + 2A - 60^\circ + 2A)]$$

$$= -2 \sin 2A [\cos 120^\circ - \cos 4A]$$

$$= -2 \sin 2A [\cos(180^\circ - 60^\circ) - \cos 4A]$$

$$= -2 \sin 2A [-\cos 60^\circ - \cos 4A]$$

$$= -2 \sin 2A [-\frac{1}{2} - \cos 4A]$$

$$= \sin 2A + 2 \cos 4A \sin 2A$$

$$= \sin 2A + \sin 6A - \sin 2A \quad (\because 2 \cos \theta \sin \theta = \sin 2\theta)$$

$$= \sin 6A = \text{R.H.S}$$

8. If $\tan \theta = \frac{1}{2}$ then p.T. $7\cos 2\theta + 8\sin 2\theta = \frac{53}{5}$

L.H.S = $7\cos 2\theta + 8\sin 2\theta$

$$= 7 \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) + 8 \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right)$$

$$= 7 \left(\frac{1 - (\frac{1}{2})^2}{1 + (\frac{1}{2})^2} \right) + 8 \left(\frac{2 \cdot \frac{1}{2}}{1 + (\frac{1}{2})^2} \right)$$

$$= 7 \left(\frac{1 - \frac{1}{4}}{1 + \frac{1}{4}} \right) + 8 \left(\frac{1}{1 + \frac{1}{4}} \right)$$

$$= 7 \left(\frac{\frac{4-1}{4}}{\frac{4+1}{4}} \right) + 8 \left(\frac{4}{4+1} \right)$$

$$= 7 \left(\frac{3}{5} \right) + 8 \left(\frac{4}{5} \right)$$

$$= \frac{21 + 32}{5}$$

$$= \frac{53}{5} = \text{R.H.S.}$$

9. p.T. $\cos 20^\circ + \cos 60^\circ + \cos 100^\circ + \cos 140^\circ = \frac{1}{2}$

L.H.S = $\cos 20^\circ + \cos 60^\circ + \cos 100^\circ + \cos 140^\circ$

$$= \cos 60^\circ + (\cos 140^\circ + \cos 100^\circ) + \cos 20^\circ$$

$$= \frac{1}{2} + 2 \cos \left(\frac{140^\circ + 100^\circ}{2} \right) \cos \left(\frac{140^\circ - 100^\circ}{2} \right) + \cos 20^\circ$$

$$= \frac{1}{2} + 2 \cos 120^\circ \cos 20^\circ + \cos 20^\circ$$

$$= \frac{1}{2} + 2 \cos (180^\circ - 60^\circ) \cos 20^\circ + \cos 20^\circ$$

$$= \frac{1}{2} + 2(-\cos 60^\circ)(\cos 20^\circ + \cos 20^\circ)$$

$$= \frac{1}{2} + 2 \cdot \left(-\frac{1}{2}\right)(\cos 20^\circ + \cos 20^\circ)$$

$$= \frac{1}{2} - \cos 20^\circ + \cos 20^\circ$$

$$= \frac{1}{2} = \text{R.H.S.}$$

10. Prove that $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{16}$

$$\text{L.H.S} = \sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$$

$$= -\frac{1}{2} \sin 10^\circ \cdot \frac{1}{2} [-2 \sin 70^\circ \sin 50^\circ]$$

$$= -\frac{1}{4} \sin 10^\circ [\cos 120^\circ - \cos 20^\circ] \quad (\because -2ss = c - c)$$

$$= -\frac{1}{4} \sin 10^\circ [\cos(180^\circ - 60^\circ) - \cos 20^\circ]$$

$$= -\frac{1}{4} \sin 10^\circ [-\cos 60^\circ - \cos 20^\circ]$$

$$= -\frac{1}{4} \sin 10^\circ \left[-\frac{1}{2} - \cos 20^\circ\right]$$

$$= \frac{1}{8} \sin 10^\circ + \frac{1}{4} \cos 20^\circ \sin 10^\circ$$

$$= \frac{1}{8} \sin 10^\circ + \frac{1}{4} \cdot \frac{1}{2} (2 \cos 20^\circ \sin 10^\circ)$$

$$= \frac{1}{8} \sin 10^\circ + \frac{1}{8} [\sin 30^\circ - \sin 10^\circ] \quad (\because 2cs = s - s)$$

$$= \frac{1}{8} \sin 10^\circ + \frac{1}{8} \left[\frac{1}{2} - \sin 10^\circ\right]$$

$$= \frac{1}{8} \sin 10^\circ + \frac{1}{16} - \frac{1}{8} \sin 10^\circ$$

$$= \frac{1}{16} = \text{R.H.S.}$$

11. Prove that $\tan^{-1}\left(\frac{3}{4}\right) + \sin^{-1}\left(\frac{4}{5}\right) = \frac{\pi}{2}$.

$$\text{L.H.S} = \tan^{-1}\left(\frac{3}{4}\right) + \sin^{-1}\left(\frac{4}{5}\right)$$

$$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{4/5}{\sqrt{1-16/25}}\right)$$

$$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{4/5}{\sqrt{9/25}}\right)$$

$$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{4/5}{3/5}\right)$$

$$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{4}{3}\right)$$

$$= \frac{\pi}{2} \quad \left(\because x \cdot y = \frac{3}{4} \cdot \frac{4}{3} = 1 \right)$$

$$= \text{R.H.S}$$

12. p.T. $\sin^{-1}\left(\frac{12}{13}\right) + \cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{63}{13}\right) = \pi$

$$\text{L.H.S} = \sin^{-1}\left(\frac{12}{13}\right) + \cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{63}{13}\right)$$

$$= \tan^{-1}\left(\frac{12/13}{\sqrt{1-\frac{144}{169}}}\right) + \tan^{-1}\left(\frac{\sqrt{1-16/25}}{4/5}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$= \tan^{-1}\left(\frac{12/13}{\sqrt{25/169}}\right) + \tan^{-1}\left(\frac{\sqrt{9/25}}{4/5}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$= \tan^{-1}\left(\frac{12/13}{5/13}\right) + \tan^{-1}\left(\frac{3/5}{4/5}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$= \tan^{-1}\left(\frac{12}{5}\right) + \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$= \pi + \tan^{-1} \left(\frac{\frac{12}{5} + \frac{3}{4}}{1 - \frac{12}{5} \cdot \frac{3}{4}} \right) + \tan^{-1} \left(\frac{63}{16} \right) \quad \left(\because x \cdot y = \frac{12}{5} \cdot \frac{3}{4} = \frac{36}{20} > 1 \right)$$

$$= \pi + \tan^{-1} \left(\frac{\frac{48+15}{20}}{\frac{20-36}{20}} \right) + \tan^{-1} \left(\frac{63}{16} \right)$$

$$= \pi + \tan^{-1} \left(\frac{63/20}{-16/20} \right) + \tan^{-1} \left(\frac{63}{16} \right)$$

$$= \pi + \tan^{-1} \left(-\frac{63}{16} \right) + \tan^{-1} \frac{63}{16}$$

$$= \pi - \tan^{-1} \frac{63}{16} + \tan^{-1} \frac{63}{16} \quad \left(\because \tan^{-1}(-x) = -\tan^{-1}(x) \right)$$

$$= \pi$$

$$\geq \text{R.H.S.}$$

13. Draw the graph of $y = \sin x$, $0 \leq x \leq 2\pi$

14. Draw the graph of $y = \cos x$, $0 \leq x \leq 2\pi$

15. Draw the graph of $y = \sin x$, $-\pi \leq x \leq \pi$

16. Draw the graph of $y = 3\cos\left(\frac{x}{2}\right)$, $0 \leq x \leq 2\pi$